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Morphological operations based on OpenCV using Erosion technique.

AIM:- Morphological operations based on OpenCV using Erosion technique.

PROGRAM:-

```
import cv2

import numpy as np

# --- Step 1: Load the image ---
img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

    exit()

# --- Step 2: Convert to grayscale and threshold ---
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)

# --- Step 3: Define structuring element (kernel) ---
kernel = np.ones((5, 5), np.uint8) # 5x5 square kernel

# --- Step 4: Apply erosion ---
eroded = cv2.erode(binary, kernel, iterations=1)

# --- Step 5: Show and save results ---
cv2.imshow("Original", img)

cv2.imshow("Binary", binary)

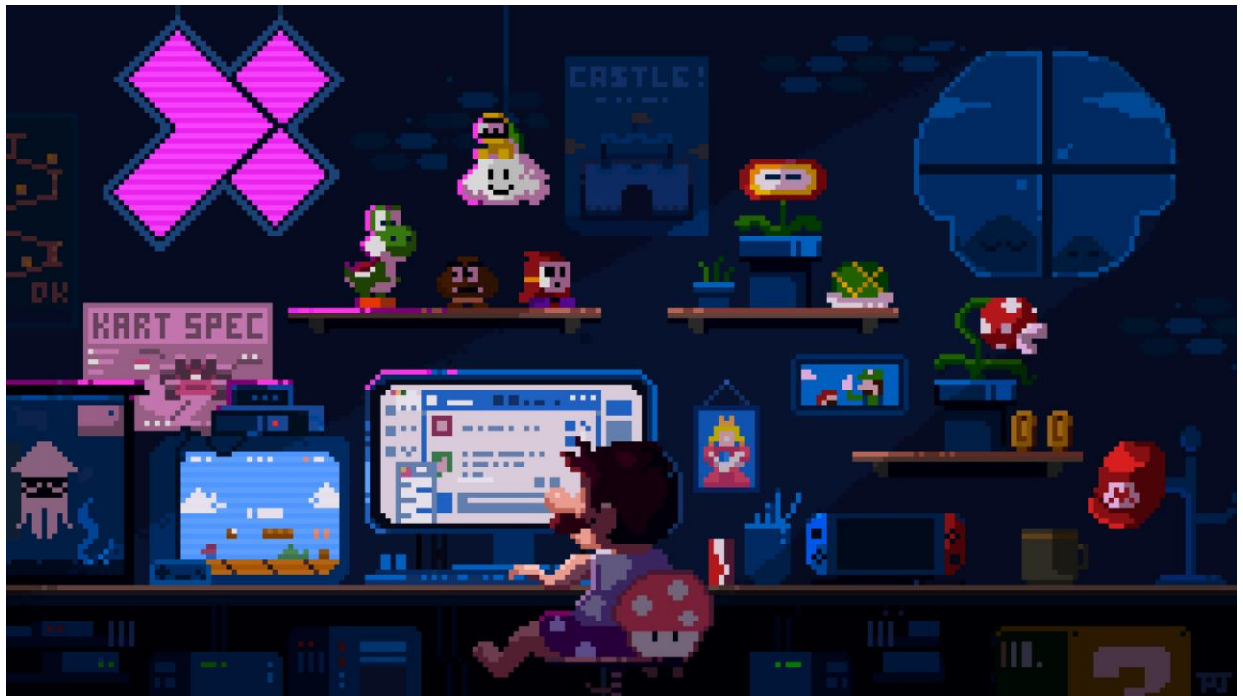
cv2.imshow("Eroded Image", eroded)

cv2.imwrite("Eroded_Image.jpg", eroded)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

INPUT:



OUTPUT:





RESULT:- Thus the program is executed successfully.